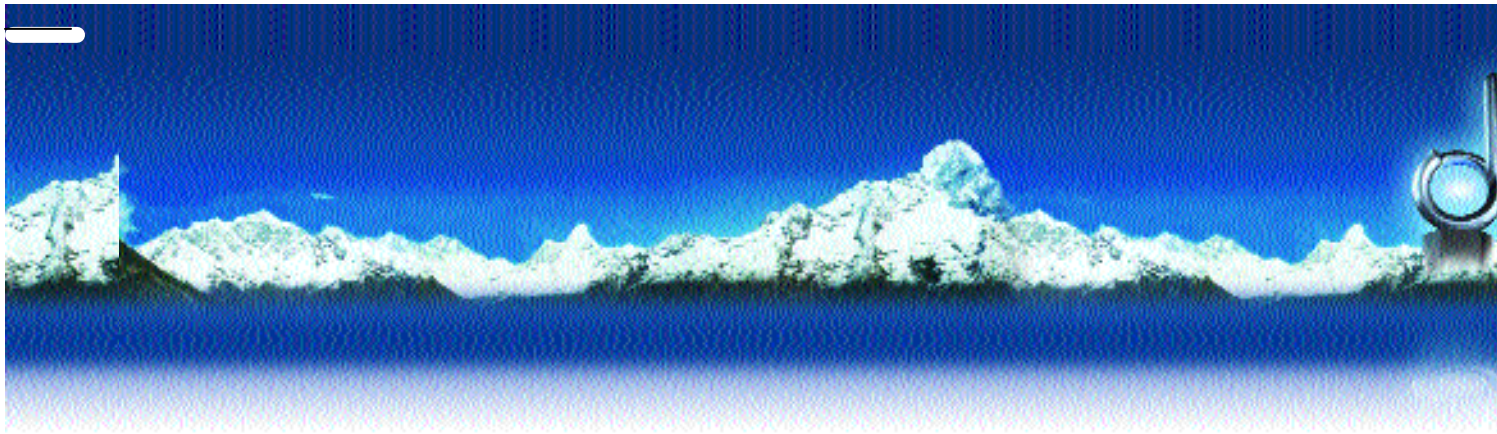
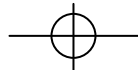




HARD DISK DRIVES

Over the past year, hard disk drives have evolved from the entry-level standard of 5,400-rpm to 7,200-rpm and capacities now dictate that hard disks of 40 GB are the bare minimum for any application. With further enhancements in fabrication processes, manufacturing costs have been lowered and this has resulted in a drop in prices. This is the primary reason why some 7,200-rpm drives today are only marginally more expensive than their 5,400-rpm counterparts.

On the right track

When looking for a hard disk drive, the best way to decide which one is ideal for you is to evaluate the technology used, along with the capacity it offers. The parameters that influence the performance are the data transfer rate while reading and writing different types of data, access time, amount of buffer memory and finally the interface.

The 5,400-rpm drives naturally deliver lower data transfer rates compared to the 7,200-rpm drives. For example, 5,400-rpm drives generally offer sequential read data transfer rates in the region of 32 to 42 MBps, while 7,200-rpm drives offer 48 to 54 MBps. The access times specified for 5,400-rpm drives should not be more than 12 ms, while for 7,200-rpm drives it should not exceed 9 ms. Faster access times will result in speedier access to distributed sets of data across your hard disk.

Today's hard disks sport ATA/100 interface and the newer ones are just beginning to feature ATA/133 interfaces. Both these interface feature maximum data transfer rates of 100 MBps and 133 MBps, respectively. However, in order to utilise them to their



**Western Digital
WD1000**

ACI Infocom,
Ph: 022-28266031

The verdict: Boasting of a 7,200-rpm spindle speed and a 100 GB storage capacity, this hard disk drive shattered all records for data transfer speed over various applications



Nominees

Name	Contacts	Comments
IBM IC35I080A V VA07-0	Wales Technologies, Ph: 022-23828100	This hard disk has a 2 MB buffer and a 7,200-rpm spindle speed. It is suited to applications that require fast response from the storage subsystem.
Western Digital WD2000	ACI Infocom, Ph: 022-28266031	Sporting a whopping 200 GB capacity, this hard disk delivered astonishing performance in various types of data transfer applications.

maximum potential, they have to be supported by a fast motherboard and the resultant speed gains are only noticed when the rotations speed of the hard disks are higher. In 7,200-rpm drives use these interfaces more efficiently than 5,400-rpm drives. Finally, the buffer on the hard disk plays an important role in holding information temporarily while data is being written to or read from the hard disk—the larger the buffer, the faster is the response of the hard disk. Nearly all hard disks available today have a 2 MB data buffer.

And the award goes to...

The drive that has the honour of being the fastest drive of the year is the Western Digital WD1000. This 100 GB hard disk is based on a 7,200-rpm spindle speed. It blazed through various tests and clocked some of the highest data transfer rates we've ever seen in an IDE hard disk drive. With its high data transfer rates in both sequential and random reads and writes, it is ideally suited to applications that demand the best in storage performance.

OPERATING SYSTEMS

Windows XP Professional

A proven operating system in a swanky new dressing, Windows XP Professional still rules with its powerful features, security and added functionality in a good-looking and intuitive interface. It incorporates features such as ClearText and the integration of hardware-accelerated graphics for the GUI. The re-worked Start menu gives access to often used applications and can be customised to individual preferences. On the security front, it incorporates the robustness and stability of Windows 2000 and has a firewall for added security. With a redefined layout for applications and system settings, controlling and accessing the settings and components of the OS is easy. The process of config-

uring new peripherals and technologies is intuitive with most of the management tasks being wizard driven. This OS even works well with older programs as they can be defined to run in an environment that mimics older Windows OSes.



**Windows XP
Professional**

Microsoft Corporation

India Pvt Ltd,
Ph: 011-26292640

The verdict: With a host of features, great looks and a robust architecture, this operating system still runs even the most demanding applications

